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DATA HANDLING AND VISUVALIZATION LAB PROGRAMMS

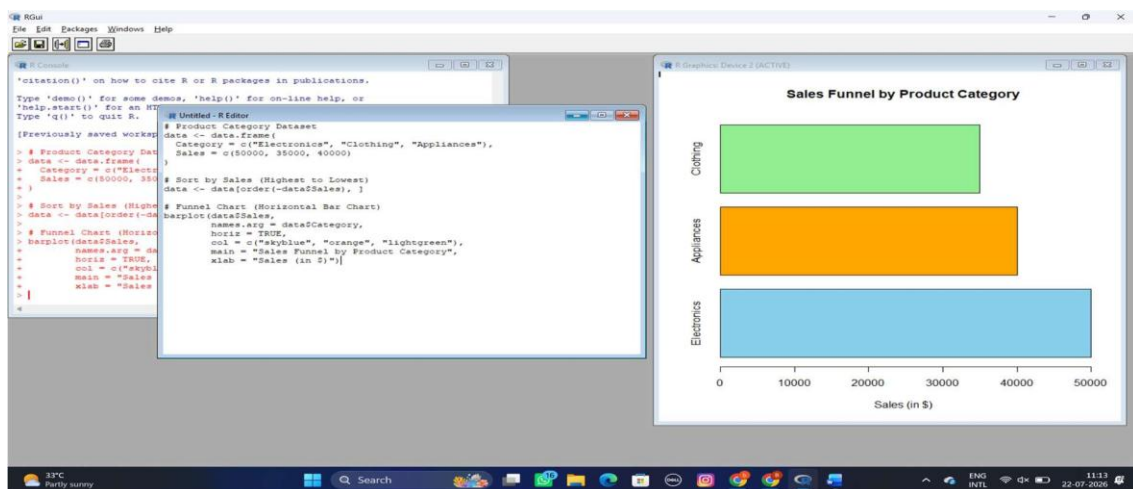
11-15:

EXPERIMENT-11:

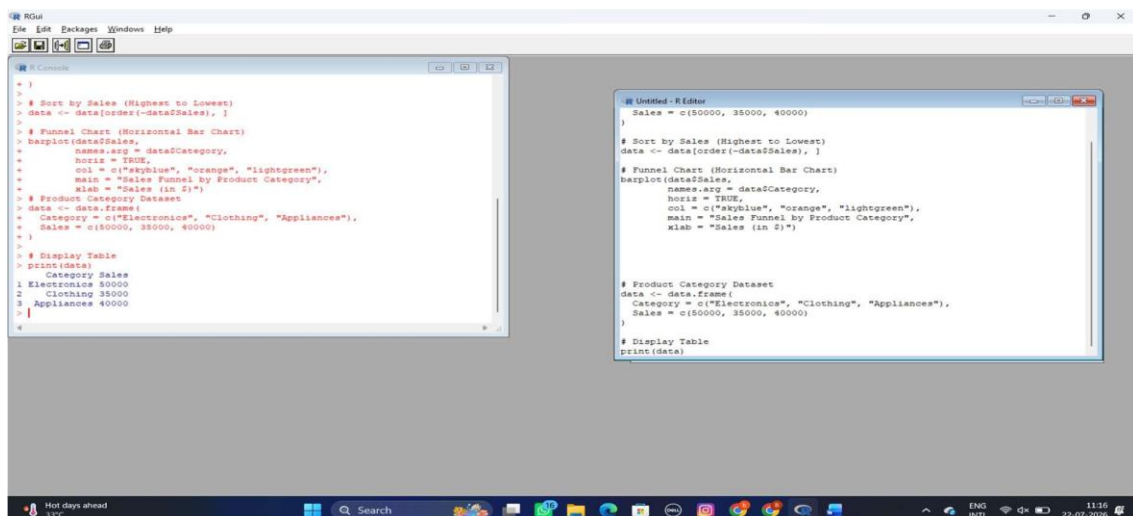
Product Category Analysis

Dataset: Product Categories

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.



2) Build a table to display the sales data for each product category. Label the table elements.



The screenshot displays the RStudio environment with the following components:

- R Console:** Shows the execution of R code to create a dataset and generate a pie chart.


```

xlab = "Sales (in $)")
# Product Category Dataset
data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
# Display Table
print(data)
# Product Category Dataset
data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
# Display Table
print(data)
# Product Category Dataset
data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
# Pie Chart
pie(data$Sales,
  labels = paste(
    col = c("skyblue", "orange", "lightgreen"),
    main = "Sales Distribution by Product Category"
  )
)

```
- Untitled - R Editor:** Contains the same R code as the console, organized into sections with comments.


```

# Product Category Dataset
data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
# Display Table
print(data)

# Product Category Dataset
data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
# Display Table
print(data)

# Product Category Dataset
data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
# Pie Chart
pie(data$Sales,
  labels = paste(
    col = c("skyblue", "orange", "lightgreen"),
    main = "Sales Distribution by Product Category"
  )
)

```
- R Graphics Device 2 (ACTIVE):** Displays a pie chart titled "Sales Distribution by Product Category". The chart is divided into three segments:
 - Electronics \$ 50000:** Represented by a light blue segment.
 - Clothing \$ 35000:** Represented by an orange segment.
 - Appliances \$ 40000:** Represented by a light green segment.

The Windows taskbar at the bottom shows the system time as 11:20 on 22-07-2026, with various background applications and system icons.

Dataset: Website Traffic

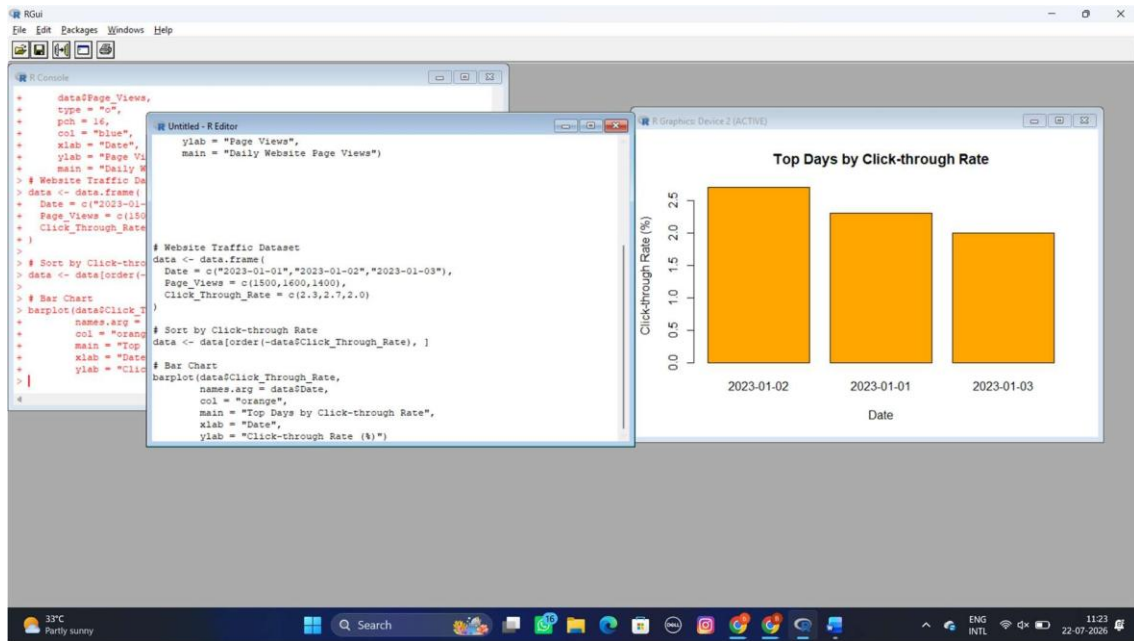
The screenshot displays the RStudio environment with several windows open:

- R Console:** Contains the following R code:

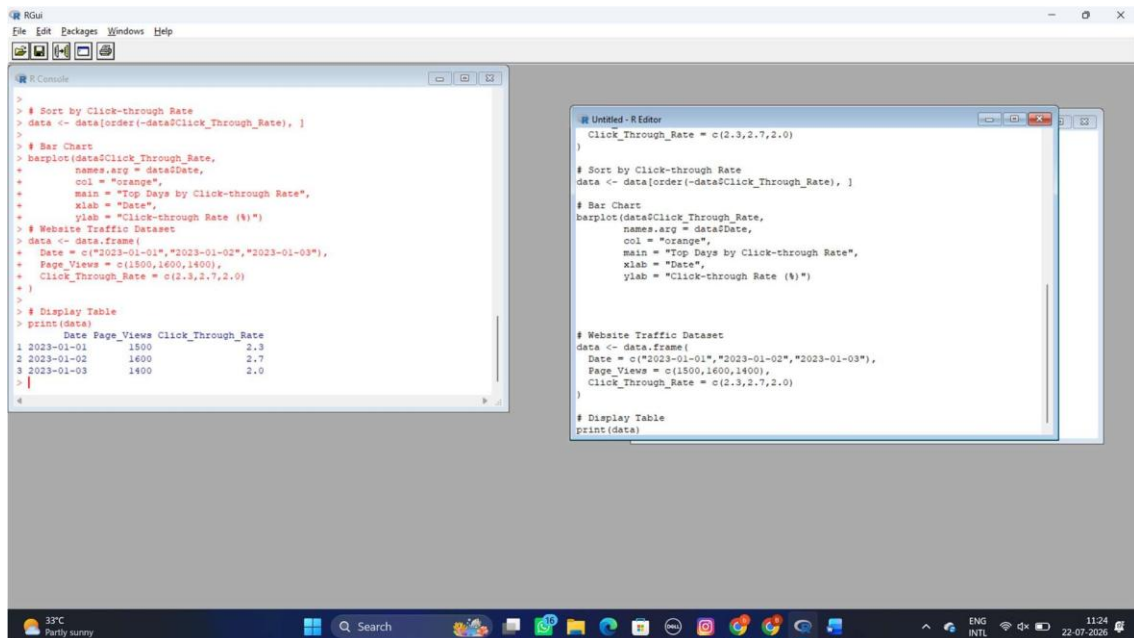

```
Sales = c(50000, 35000, 40000)
# }
# Pie Chart
pie(data$Sales,
  labels = paste(
    col = c("skyblue",
    main = "Sales Data")
# Website Traffic Data
data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  Click_Through_Rate = c(2.3, 2.7, 2.0)
)
# Line Chart
plot(data$Date,
  data$Page_Views,
  type = "o",
  pch = 16,
  col = "blue",
  xlab = "Date",
  ylab = "Page Views",
  main = "Daily Website Page Views")
# }
# Line Chart
plot(data$Date,
  data$Page_Views,
  type = "o",
  pch = 16,
  col = "blue",
  xlab = "Date",
  ylab = "Page Views",
  main = "Daily Website Page Views")
# }
```
- Untitled - R Editor:** Contains the same R code as the console, with comments indicating the purpose of each section: "Website Traffic Dataset", "Pie Chart", "Website Traffic Data", and "Line Chart".
- R Graphics Device 2 (ACTIVE):** Displays a line plot titled "Daily Website Page Views". The x-axis is labeled "Date" and has ticks for "Jan 01", "Jan 02", and "Jan 03". The y-axis is labeled "Page Views" and ranges from 1400 to 1600. The plot shows a blue line with circular markers at each data point: (Jan 01, 1500), (Jan 02, 1600), and (Jan 03, 1400).

The Windows taskbar at the bottom shows the system date and time as 11:22 on 22-07-2025, along with various icons for applications and system status.

2. Generate a bar chart showing the top N days with the highest clickthrough rates. Label the chart elements.



3. Build a table to show the website traffic data for each day. Label the table elements.

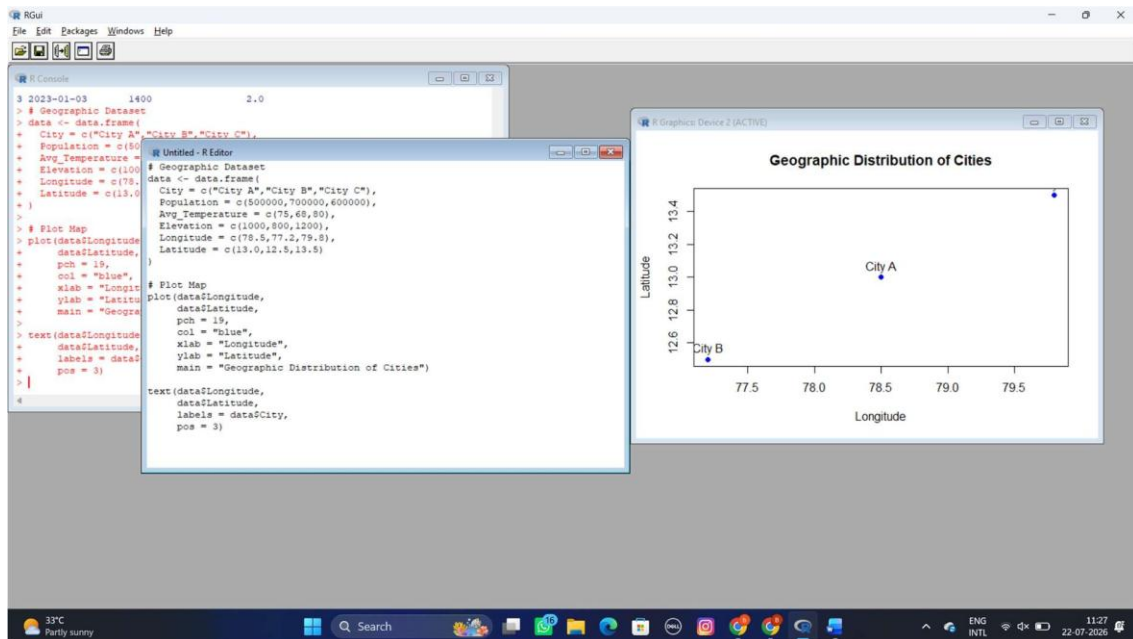


EXPERIMENT 13:

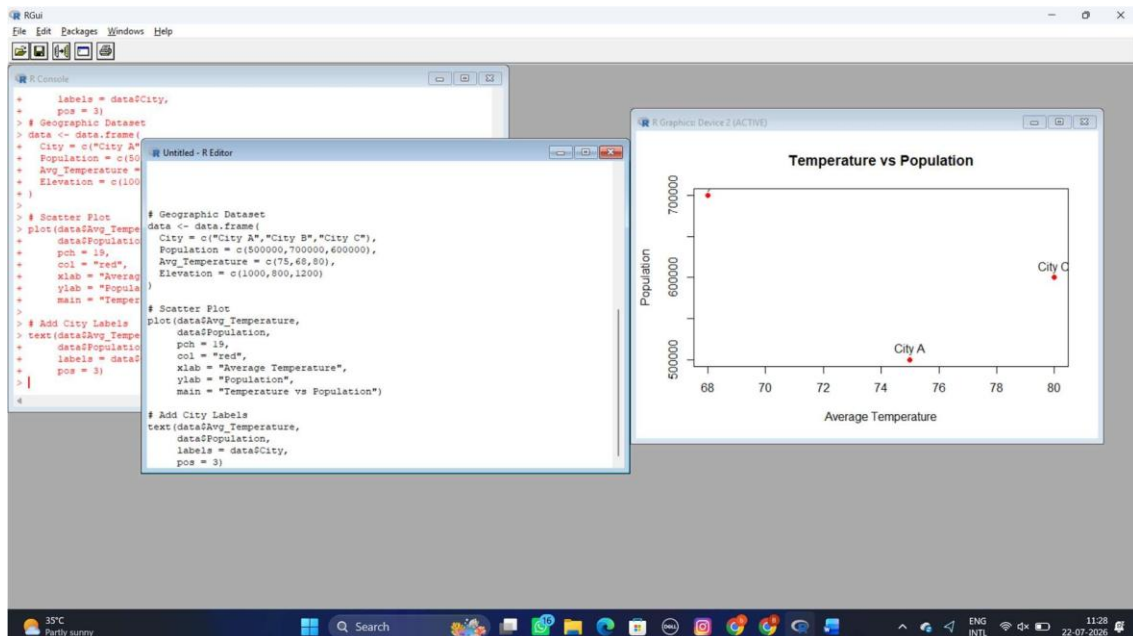
Dataset: Geographic Data

City Population Avg. Temperature Elevation

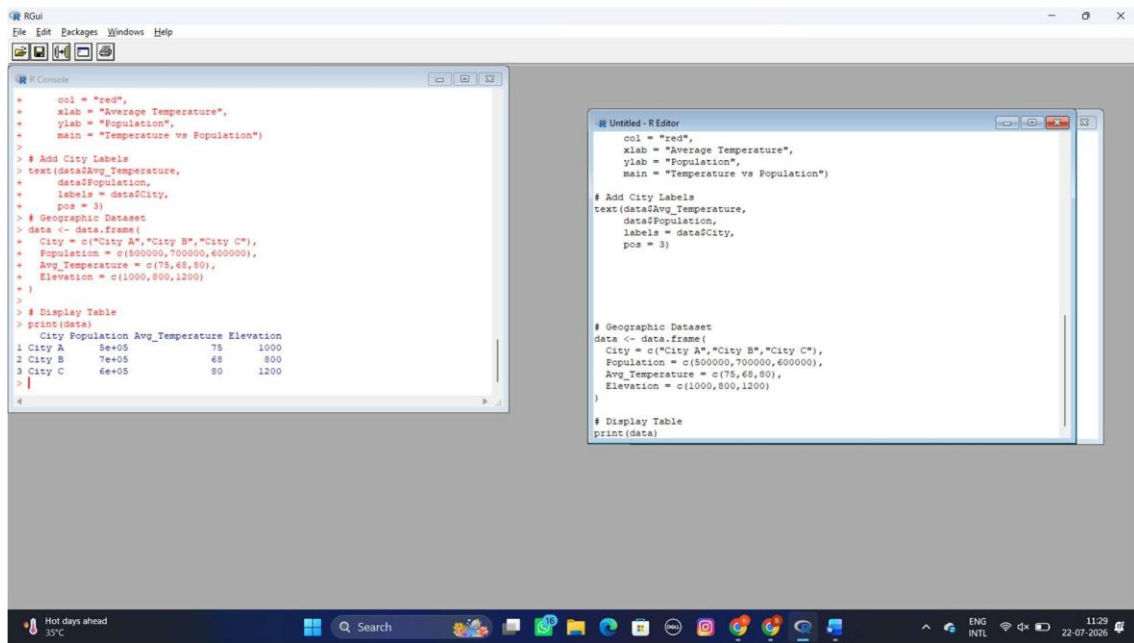
1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.



2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.



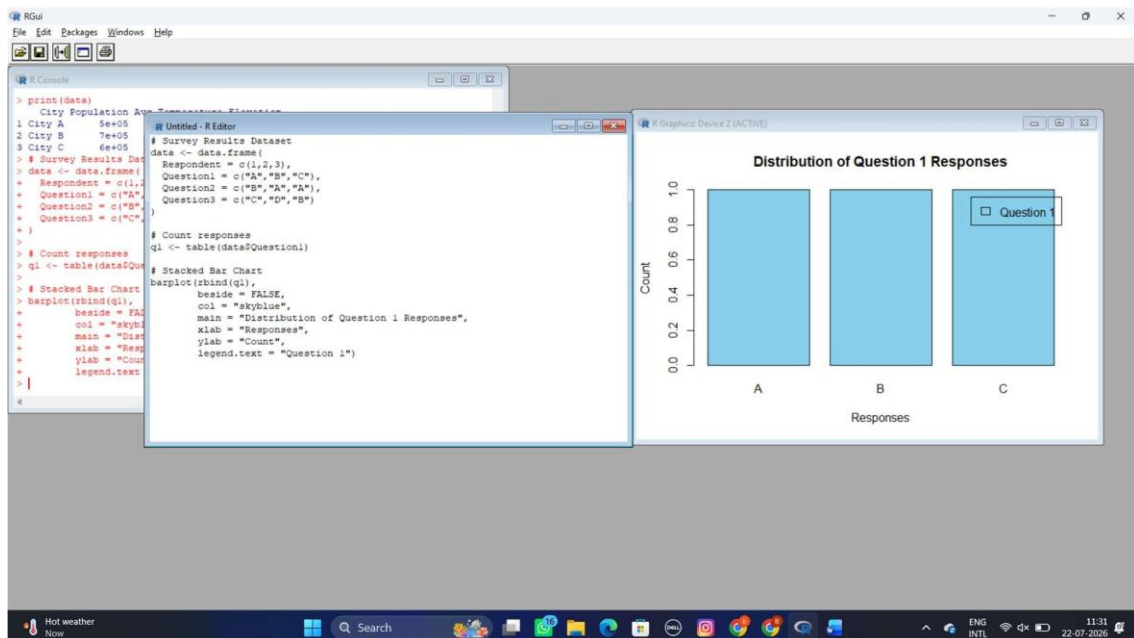
3. Build a table to display the geographic data for each city. Label the table elements.



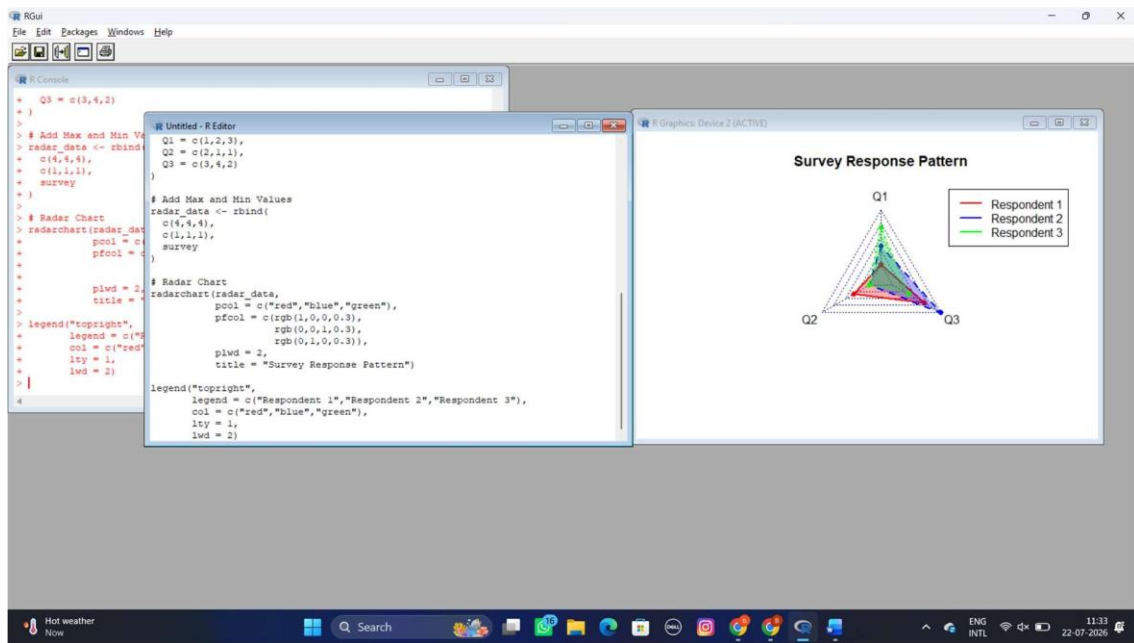
EXPERIMENT 14:

14. Dataset: Survey Results

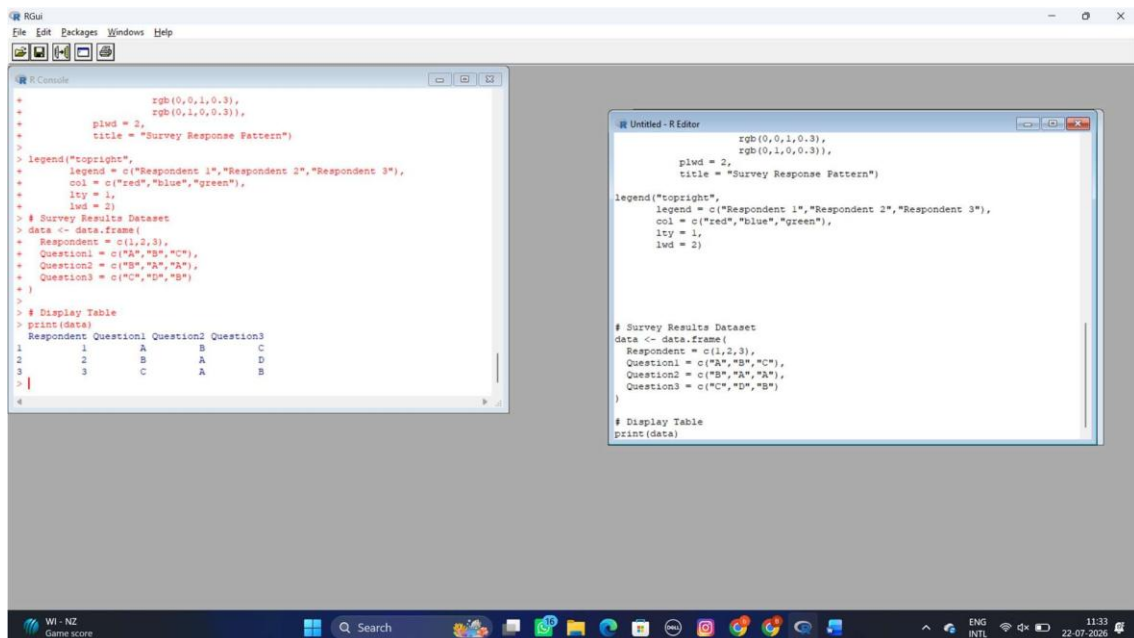
1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.



2. Generate a radar chart to visualize the overall pattern of responses across all three questions.



3. Build a table to show the survey response data for each respondent.
Label the table elements.

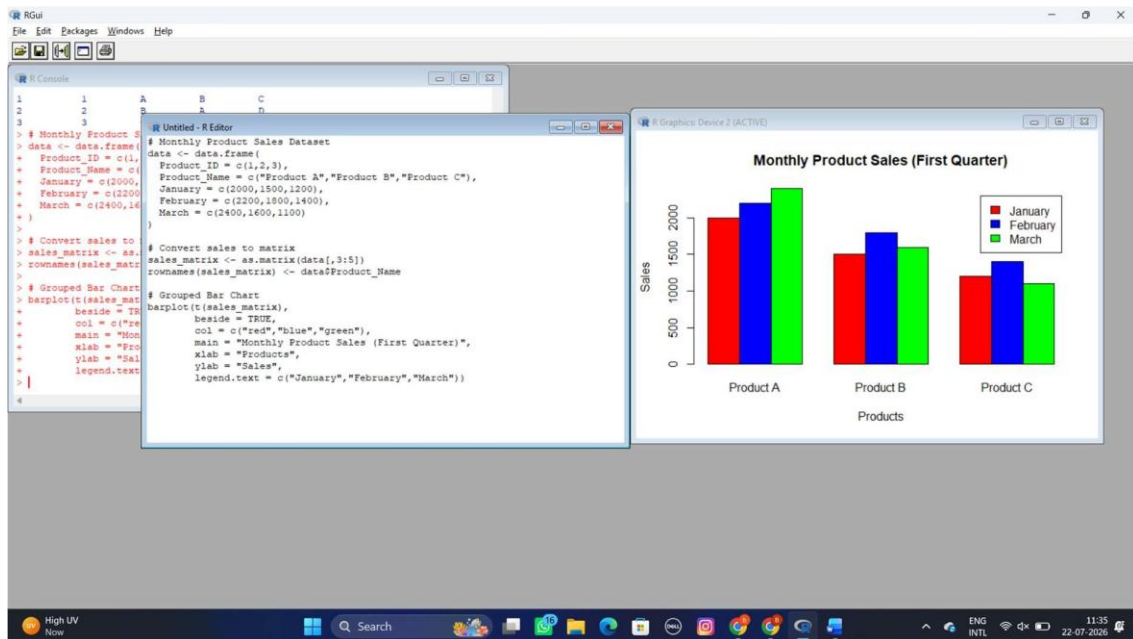


EXPERIMENT 15:

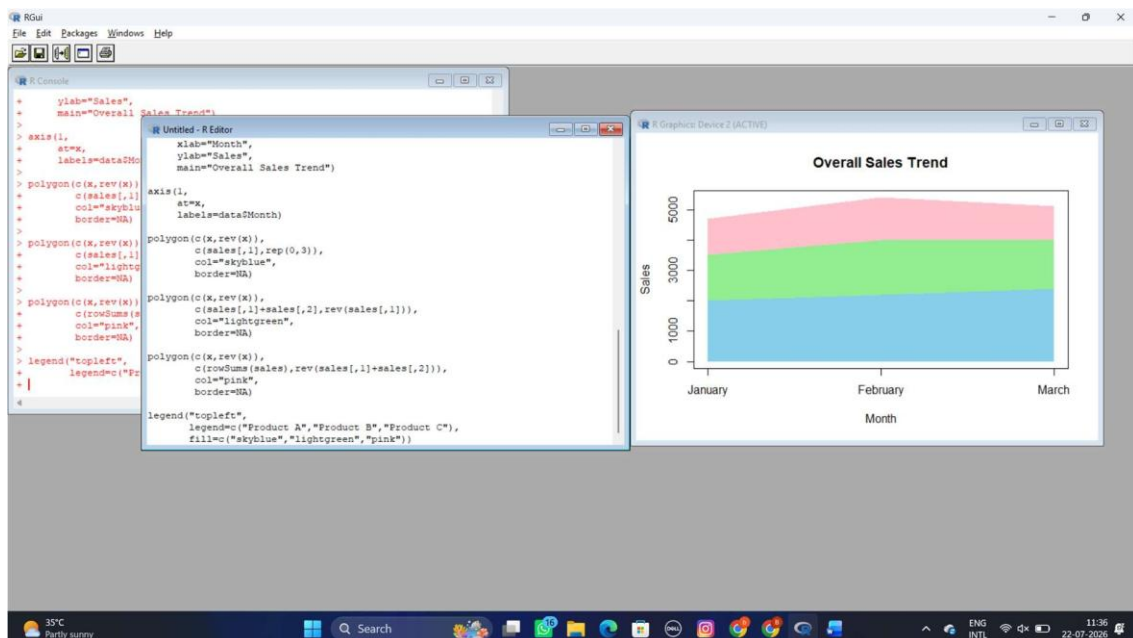
Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.



3. Build a table to show the monthly sales data for each product. Label the table elements.

